A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, III: The division-algebra side and its local character transform

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Abstract

Let D/F be a central division algebra of prime degree r and $\widetilde{D^\times}$ the n-fold cover (n odd) specified in Part I, where $(n, r) = 1$. We construct and normalize a unitary transform L_D of genuine and ordinary invariant class functions, with local coefficient $|n|_F^{-(r-1)/2}$. When the residue characteristic is prime to n, a unique splitting over the reduced-norm-one group, together with an Adams operation, proves that L_D bijects irreducible characters and preserves dimensions. At places dividing n we obtain a nonsplitting and divisibility obstruction. We give the normalized orbital-integral transfer and its trace-pairing identity. Globally we prove an almost-everywhere local-to-global principle for nth powers modulo scalars, and explain the relevant rational-orbit and measure considerations from Kottwitz's work. The tame character relation with the cover of GL_r follows, within its published correspondence range, by an explicit cocycle, star-map, central-character and root-sum check. The wild-place character correspondence and the adelic trace-formula comparison remain open.

Contents

1 Introduction and results **1**

2 Local setting and good elements **2**

3 Power maps on elliptic tori **4**

4 The normalized local class-function transform **5**

5 Irreducible characters and the integrality obstruction **6**

6 Adams operations and finite-group characters **8**

7 The tame case: splitting and irreducible character integrality **9**

8 Places dividing n: non-splitting and constraints **14**

9 Local orbital-integral transfer on $D^\times$ **15**

9.1 Test functions and orbital integrals 15

9.2 An explicit transfer theorem 16

9.3 The duality with character transfer 17

10 Kottwitz's framework for rational orbits and inner forms **18**

11 Global power classes and a Hasse principle **19**

12 The global comparison: remaining problems **21**
12.1 Four proof obligations 21
12.2 Conditional representation-theoretic consequences 22

13 Selected terminology and notation **22**

1 Introduction and results

The classical local Jacquet–Langlands correspondence identifies irreducible representations of $D^\times$ with essentially square-integrable representations of $\mathrm{GL}_r(F)$ in the appropriate central-character families [9]. It does not produce a correspondence between ordinary representations of $D^\times$ and genuine representations of its metaplectic cover: pullback from the linear group is not genuine. This paper studies the division-algebra side directly and provides local results independent of a conjectural global trace-formula comparison.

Position within the series. Part I constructs the specified covers, computes their good elements, and proves matching for regular-supported functions; Part II proves matching of arbitrary functions (its Theorem E), identifies the germs at the center, and proves an improved global transfer (its Theorem C′). Here the central object is the transform L_D on the compact-mod-center group $D^\times$. The principal issues and their status are:

Topic	**Assertion**	**Status**
Covering theory	Scalar centrality and good-locus description from Part I	[E]
Local harmonic analysis	Unitary L_D and explicit coefficient b_D	[P]
Tame representations	Splitting over D^1 and irreducible-character bijection for $p \nmid n$	[P]
Wild representations	Nonsplitting and necessary divisibility constraints	[P]
	Integrality of L_D at places dividing n	[C]
Character relation (D2) of Part I	Tame places, with the Flicker–Kazhdan–Zou theorem available (in particular when $(n, N_r) = 1$); cocycle and star-map compatibility checked below (Corollary 7.8)	[P]
Test functions	Normalized local orbital-integral transfer and trace adjunction	[P]
Global arithmetic	Local-to-global powers modulo scalars outside any finite set	[P]
Global comparison	Good-element matching, adelic measures, spectral separation	[C]

As in Parts I and II, **[P]** means proved here (possibly using results marked **[E]**), **[E]** means established elsewhere (including Parts I and II) and used as a black box, **[O]** means that a proof is outlined but not fully written out or checked, and **[C]** means conjectural: new mathematical input is required. These statuses concern the precise stated hypotheses; the local claims rely on the specific covering cocycle of Part I, not on an arbitrary central extension.

The two distinct maps in the paper should be kept separate: the unitary operator on invariant L^2 class functions, and a bijection of individual irreducible characters. The first exists at all division places, whereas the second is established here only when $p \nmid n$. The Adams operation provides the bridge in that case.

After the local results, we review the part of Kottwitz's arithmetic framework needed for rational conjugacy and inner forms, then prove a Hasse principle for the global power quotient. The final

section states what remains for an adelic metaplectic Jacquet–Langlands comparison. A short glossary records the specialized terminology.

The running example

In the series' running example, $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1-\zeta_3)$ and $\mathfrak{p}_7 = (3+\zeta_3)$ (Part I, §1, and `notes/mjl_running_example.tex`), the two places of S illustrate the two regimes of this paper. At $\mathfrak{p}_7$, which does not divide n, $b_D = c_D = 1$, the discriminant ratio is identically 1 (Lemma 7.4), and L_D is a dimension-preserving bijection of irreducible characters (Theorem 7.6). At $\mathfrak{p}_3$, where $K_v = \mathbb{Q}_3(\zeta_3)$ and $|3|_v = 1/9$, one has $b_D = 3$ and $c_D = 27$. There the cover does not split, and genuine irreducibles have dimension divisible by 3 (Proposition 8.1). Integrality of L_D would force $L_D\mathbf{1}$ to be irreducible of dimension 27 (Remark 5.5). Whether it is remains open; see Part IV.

2 Local setting and good elements

We fix the local field, central-character conventions, and the covering-theoretic theorem imported from Part I. The distinction between a good element of the cover and an ordinary regular element is essential to the definition of the transform. Let F be a finite extension of $\mathbb{Q}_p$ containing μ_n, where $n \geq 1$ is odd. Let $r \geq 2$ be prime and $(n, r) = 1$. Let D/F be a central division algebra of degree r, $H = D^\times$, and $Z = F^\times$ its scalar center. We use the normalized local absolute value $|\cdot|_F$ ($|\varpi_F|_F = q_F^{-1}$).

Let

$$1 \longrightarrow \mu_n \longrightarrow \widetilde{H} \xrightarrow{p} H \longrightarrow 1 \tag{1}$$

be a locally compact topological central extension. Throughout we use *the specified cocycle of Part I*, rather than an arbitrary central extension. In its parameter convention, $n \mid r - 1 + 2rc_v$; under that condition the inverse image $\widetilde{Z} = p^{-1}(Z)$ is central (Part I, Lemma 2.2). The cocycle condition is essential; a general n-fold extension need not have central scalar preimage. References to Part I below refer to its stated cocycle conventions.

In the notation of Parts I and II, $\widetilde{H} = \overline{G}_D(F)$, $\widetilde{Z} = \overline{Z}$, and $\overline{G}_v$ denotes the Kazhdan–Patterson cover of $\mathrm{GL}_r(F)$ with parameter c_v; the normalized orbital integral $J = \Delta O$ of §9 is the function $F = \Delta\Phi$ of Part I, Definition 4.1, for the measures used here. Fix a unitary character $\widetilde{\theta} : \widetilde{Z} \to \mathbb{C}^\times$ whose restriction to μ_n is faithful. Write $\theta(a) = \widetilde{\theta}(a^*)$ for $a \in Z$, where the star map is defined next. For invariant functions we impose the central equivariance conventions $f(ah) = \theta(a)f(h)$ and $\widetilde{f}(zh) = \widetilde{\theta}(z)\widetilde{f}(h)$.

Definition 2.1 (The star map)**.** For $h \in H$ choose any lift $\widetilde{h} \in \widetilde{H}$ and set

$$h^* = \widetilde{h}^{\,n} \in \widetilde{H}. \tag{2}$$

The definition is independent of the lift. It is conjugation equivariant, and for $a \in Z$ and $h \in H$ one has $(ah)^* = a^*h^*$.

Proof. Another lift is $\zeta\widetilde{h}$ with $\zeta \in \mu_n$, whose nth power is $\widetilde{h}^n$. Conjugation equivariance follows by lifting the conjugating element. Elements of $\widetilde{Z}$ are central by assumption, hence $(\widetilde{a}\widetilde{h})^n = \widetilde{a}^n\widetilde{h}^n$. In particular $a \mapsto a^*$ is a homomorphism from Z to $\widetilde{Z}$. (For a maximal torus it is also a homomorphism, because n is odd; see below.) □

Every noncentral $h \in D^\times$ generates a maximal subfield $E = F(h)$, of degree r. Set $T = E^\times$; it is its own centralizer in H. A regular element of H means a noncentral element. The corresponding Weyl discriminant is

$$\Delta(h) = \left|\det(1 - \mathrm{Ad}(h) \mid \mathfrak{h}/\mathfrak{t})\right|_F^{1/2} = \prod_{i<j} \left| \frac{(\lambda_i - \lambda_j)^2}{\lambda_i \lambda_j} \right|_F^{1/2}, \tag{3}$$

where $\lambda_1, \ldots, \lambda_r$ are the conjugate eigenvalues in a splitting field, and the product is interpreted with the normalized absolute value.

Definition 2.2 (Good regular element)**.** A regular $\widetilde{h} \in \widetilde{H}$ is good if it commutes with $p^{-1}(Z_H(p(\widetilde{h})))$. The definition is equivalent to the usual statement that the commutator character on the preimage of the regular centralizer is trivial. Every h^* is good, since its commutators are nth powers in μ_n.

Remark 2.3. If $\widetilde{h}$ is regular and not good, every genuine conjugation-invariant function $\widetilde{f}$ vanishes at $\widetilde{h}$: some lift $\widetilde{t}$ centralizing $p(\widetilde{h})$ has $\widetilde{t}\widetilde{h}\widetilde{t}^{-1} = \zeta\widetilde{h}$, with $\zeta \neq 1$; invariance and genuine central equivariance then force $\widetilde{f}(\widetilde{h}) = 0$.

Proposition 2.4 ([**E**] Good lifts on the division side (Part I))**.** *For each maximal subfield E/F of D, the good elements of $p^{-1}(E^\times)$ are precisely $z\,t^*$ with $z \in \widetilde{Z}$ and $t \in E^\times$. Equivalently, modulo $\widetilde{Z}$ the good subset corresponds to $(E^\times/F^\times)^n$. These are the good-element and commutator results established in Part I, Lemma 2.2 and Proposition 3.5(a); the local index and preservation of regularity are treated there in Proposition 3.5(b),(d).*

3 Power maps on elliptic tori

The local index of the power map and the ratio of Weyl discriminants control the normalization of L_D. Both calculations use the coprimality of n and r, including at wild places.

Lemma 3.1 ([**P**] No new singular elements)**.** *If $h \in H$ is noncentral, h^n is noncentral. In particular the image of a regular element under the star map is regular.*

Proof. Suppose $h^n \in F^\times$. The field $L = F(h)$ is an intermediate field of a maximal subfield of D, hence $[L : F]$ divides r. Because F contains μ_n and $h^n \in F^\times$, Kummer theory shows $[L : F]$ divides n. The coprimality $(n, r) = 1$ therefore gives $[L : F] = 1$, contrary to the assumption that h is noncentral. □

Lemma 3.2 ([**P**] Degree of the power map)**.** *For any degree-r extension E/F embedded in D, the map*

$$P_n : E^\times/F^\times \longrightarrow E^\times/F^\times, \qquad [t] \longmapsto [t^n]$$

is injective and has open image of index

$$I_F(n, r) = \left[E^\times/F^\times : (E^\times/F^\times)^n\right] = |n|_F^{-(r-1)}. \tag{4}$$

The index is independent of E.

Proof. If $t^n \in F^\times$, Kummer theory over a field containing μ_n gives $[F(t) : F] \mid n$. Since $F(t) \subset E$ and $[E : F] = r$, this degree also divides r; hence $t \in F^\times$, proving injectivity.

The quotient $A = E^\times/F^\times$ is a compact abelian p-adic Lie group. Its p-adic analytic dimension is $d = ([E : \mathbb{Q}_p] - [F : \mathbb{Q}_p]) = (r - 1)[F : \mathbb{Q}_p]$. The structure theorem for compact abelian p-adic

Lie groups writes $A \simeq \mathbb{Z}_p^d \times C$ for a finite abelian group C. The injectivity already proved implies $A[n] = \{1\}$, so multiplication by n is an automorphism of C. On $\mathbb{Z}_p^d$ it has index $p^{v_p(n)d}$. Since $|n|_F = p^{-[F:\mathbb{Q}_p]v_p(n)}$, this is exactly $|n|_F^{-(r-1)}$. (The argument also handles $p \mid n$.) □

Lemma 3.3 ([**P**] Discriminant ratio)**.** *For noncentral h, the ratio*

$$R(h) = \frac{\Delta(h)}{\Delta(h^n)}$$

is locally constant and nowhere zero. It extends uniquely to a locally constant function on all of H, taking the value

$$R(a) = |n|_F^{-r(r-1)/2} \qquad (a \in F^\times). \tag{5}$$

Proof. By Lemma 3.1, both discriminants are nonzero at every noncentral element. They are continuous functions with values in the discrete multiplicative group $q_F^{\frac{1}{2}\mathbb{Z}}$ and hence locally constant on this open set.

For h sufficiently close to a scalar $a \in F^\times$, all its eigenvalues are close to a in a finite splitting extension. Factoring

$$\lambda_i^n - \lambda_j^n = (\lambda_i - \lambda_j) \sum_{k=0}^{n-1} \lambda_i^{n-1-k} \lambda_j^k,$$

the final factor approaches na^{n-1} and thus has constant absolute value $|na^{n-1}|$ for h in a sufficiently small neighborhood. The factors involving $\lambda_i \lambda_j$ in the normalized discriminant cancel the $|a|^{n-1}$ powers. Each of the $r(r-1)/2$ unordered pairs consequently contributes $|n|_F^{-1}$ to $R(h)$. This proves the formula and locally constant extension; uniqueness follows from density of the regular set. □

4 The normalized local class-function transform

We first transfer *invariant functions*; the problem of sending individual irreducible characters to irreducible characters is addressed separately. The transform is normalized using quotient probability measures. We equip H/Z with Haar probability measure. Because $\widetilde{Z} = p^{-1}(Z)$, the quotient $\widetilde{H}/\widetilde{Z}$ maps isomorphically to H/Z; we transport the probability measure through this isomorphism. Each elliptic torus quotient T/Z likewise has its probability Haar measure. These conventions are the relevant ones for *local unitarity*; matching geometric distributions globally may require additional, separately stated orbital-integral conventions.

Write $\mathcal{H}_\theta(H)$ for the Hilbert space of L^2 conjugation-invariant functions on H with θ-equivariance under Z. Similarly, write $\mathcal{H}_{\widetilde{\theta}}(\widetilde{H})$ for the genuine invariant functions with $\widetilde{\theta}$-equivariance under $\widetilde{Z}$. The inner products are taken over H/Z and $\widetilde{H}/\widetilde{Z}$, respectively.

Definition 4.1 (The division-side transform)**.** Using Proposition 2.4, define on regular elements

$$\boxed{(L_D\varphi)(z\,t^*) = |n|_F^{-(r-1)/2}\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\,\varphi(t), \quad z \in \widetilde{Z},\ t \in T^{\mathrm{reg}},} \tag{6}$$

and let $L_D\varphi = 0$ at nongood regular elements. The value at singular elements is understood by continuous extension whenever φ is smooth.

Lemma 4.2 ([**P**] Well-definedness and smoothness)**.** *The expression (6) is independent of the choice of z, t and of the maximal subfield $E = F(t)$. It sends locally constant invariant functions to locally constant invariant functions.*

Proof. For a good regular element, the regular centralizer is uniquely $E^\times$. If $zt^* = z'(t')^*$ then t'^n and t^n differ by a scalar. Since their centralizers are E and $t', t \in E^\times$, the injectivity in Lemma 3.2 gives $t' = at$ with $a \in F^\times$. As $(at)^* = a^*t^*$ and $\Delta(at) = \Delta(t)$, the formula changes by the same factor $\theta(a) = \widetilde{\theta}(a^*)$ on either side. Conjugation equivariance follows from Definition 2.1.

On a regular semisimple conjugacy chart, the image of the nth-power map on T/Z is an open-and-closed subgroup by Lemma 3.2. Thus the prescribed function is locally constant on both the good and nongood regular loci, using Lemma 3.3. A sufficiently small neighborhood of any central element consists of elements represented by zt^*: the local nth-root map exists by the nonarchimedean inverse function theorem, since n is nonzero in F. The same discriminant ratio has the constant value of (5) there. Hence the function extends locally constantly across the center. Since r is prime there are no other nonregular elements. □

Theorem 4.3 ([**P**] Unitarity and explicit constant)**.** *For the specified cover of Part I, L_D extends uniquely to a unitary isomorphism*

$$L_D : \mathcal{H}_\theta(H) \xrightarrow{\sim} \mathcal{H}_{\widetilde{\theta}}(\widetilde{H}). \tag{7}$$

The positive coefficient $b_D = |n|_F^{-(r-1)/2}$ in (6) is uniquely determined by unitarity for the quotient probability measures used here.

Proof. The Weyl integration formula, applied to $|\varphi|^2$, reduces the norm on H/Z to a sum over elliptic maximal tori T/Z of weighted integrals

$$\frac{1}{|W(H,T)|}\int_{T/Z} \Delta(t)^2 |\varphi(t)|^2 \, dt.$$

There is no need for a separate Weyl integration theorem on the covering group: since $\widetilde{H}/\widetilde{Z} \simeq H/Z$, the function $|\widetilde{f}|^2$ is an honest conjugation-invariant function on H/Z, and the ordinary Weyl formula applies directly. By Remark 2.3 and Proposition 2.4, a genuine invariant function vanishes off the open-and-closed subset $(T/Z)^n$ of each torus quotient.

The power map $P_n : T/Z \to (T/Z)^n$ is a topological group isomorphism. Since T/Z has probability Haar measure and $(T/Z)^n$ has index $I = |n|_F^{-(r-1)}$ in T/Z, for every integrable function Q on this subgroup,

$$\int_{(T/Z)^n} Q(x)\, dx = \frac{1}{I}\int_{T/Z} Q(t^n)\, dt. \tag{8}$$

Consequently, with a provisional coefficient $b > 0$, the covered Weyl integral equals

$$\frac{b^2}{I\,|W(H,T)|}\int_{T/Z} \Delta(t)^2 |\varphi(t)|^2 \, dt.$$

The index is independent of T, so the equality of norms for all class functions is equivalent to $b^2 = I$. This gives $b = b_D$.

To see surjectivity, start with a smooth genuine invariant function on the cover. It vanishes off the good regular set. By the uniqueness in Lemma 4.2, the formula defining L_D can be inverted on every good regular element. The inverse has the same local smoothness properties, by the inverse of P_n on its open image. This proves surjectivity on a dense subspace and therefore the unitary isomorphism on completions. □

Remark 4.4 (Local versus global normalization; comparison with Zou)**.** The coefficient $b_D = |n|_F^{-(r-1)/2}$ is determined by the power-map Jacobian and the quotient probability measures in Theorem 4.3. A common compatible rescaling of Haar measures does not alter the ratio, but the proposed global trace formula requires independent, explicit adelic normalizations.

Zou's corrected local Flicker–Kazhdan character relation [7, (3.2) and Remark 3.3] has elliptic coefficient

$$\frac{n}{d_F(d_r)|n|_F^{r/2}}, \qquad d_F(m) = m|m|_F^{-1/2}, \qquad d_r = \gcd(n, 2rc + r - 1).$$

Under Part I's scalar-centrality congruence one has $d_r = n$, and this coefficient simplifies algebraically to b_D. To identify the *character transforms*, however, one must also identify the two cocycles, their star maps, the central-character conventions, and the discriminant and orbital-measure normalizations. The equality of numerical coefficients alone is not such an identification; see Corollary 7.8.

5 Irreducible characters and the integrality obstruction

The unitary transform has a matrix relative to irreducible characters, but unitarity alone does not force that matrix to be integral. We record explicit arithmetic tests that any true character correspondence must pass. Since H/Z is compact, every smooth irreducible representation with a fixed central character is finite dimensional. Its character is a locally constant invariant function on H. The same holds on the cover.

Corollary 5.1 ([**P**] A unitary character matrix)**.** *Let ρ range over $\mathrm{Irr}_\theta(H)$ and τ over $\mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})$. For each ρ there is a* finite *expansion*

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau. \tag{9}$$

The matrix $A = (a_{\rho\tau})$ is unitary (in the sense of the associated operator on ℓ^2), and its rows and columns are finite.

Proof. Schur orthogonality for compact groups modulo the fixed center gives orthonormal bases of irreducible characters for the two invariant L^2 spaces. Theorem 4.3 therefore makes A unitary. Local constancy and compactness modulo center show that each smooth invariant function is invariant under an open normal subgroup of the compact quotient, so it factors through a finite quotient (or through a corresponding finite-dimensional projective quotient with fixed central character). Its irreducible-character expansion is therefore finite. Applying this observation to the inverse transform gives finite columns as well. □

Corollary 5.2 ([**P**] What integral coefficients would imply)**.** *Suppose the coefficients $a_{\rho\tau}$ in (9) are integers for every ρ, τ. Then L_D maps each irreducible character to a unique irreducible character with positive sign, and it induces a bijection*

$$\mathrm{Irr}_\theta(H) \xrightarrow{\sim} \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H}).$$

More precisely, if $L_D\Theta_\rho = \Theta_\tau$, then

$$\boxed{\dim \tau = |n|_F^{-(r^2-1)/2} \dim \rho.} \tag{10}$$

Proof. An integral vector of squared ℓ^2-norm one has exactly one nonzero entry, equal to $+1$ or -1. The sign cannot be negative because $\Theta_\rho(1) = \dim \rho > 0$ and the continuous value at the identity prescribed by (6) is

$$(L_D \Theta_\rho)(1) = b_D \, |n|_F^{-r(r-1)/2} \dim \rho > 0.$$

The same argument applied to L_D^{-1} yields bijectivity. Substituting $b_D = |n|_F^{-(r-1)/2}$ gives (10). □

Remark 5.3 (A concrete wild-place check)**.** Take $r = 2$, $n = 3$, $F = \mathbb{Q}_3(\zeta_3)$. Then $|3|_F = 3^{-2} = 1/9$, so $I = 9$, $b_D = 3$, and formula (10) predicts $\dim \tau = 27 \dim \rho$ *if* integral irreducible transfer holds. This is a sharp, falsifiable numerical consequence of the proposed correspondence; it is not evidence that the correspondence is already proved.

Lemma 5.4 (**[P]** Integral local scaling constants)**.** *If $p \mid n$ and $\mu_n \subset F$, with n odd as assumed, then*

$$b_D = |n|_F^{-(r-1)/2}, \qquad c_D = |n|_F^{-(r^2-1)/2}$$

are positive integers.

Proof. Write $p^k \| n$ and $d = [F : \mathbb{Q}_p]$. Since $\mu_{p^k} \subset F$ and p is odd, $p - 1$ divides d; in particular d is even. For the normalized absolute value, $|n|_F^{-1} = p^{kd}$ is a perfect square. Both exponents above therefore give integral nonnegative powers of p. □

Remark 5.5 (A falsifiable wild-place divisibility test)**.** Suppose the matrix of Corollary 5.1 is integral at a wild place, so the transform is a bijection on irreducible characters. Every genuine $\tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})$ must then satisfy $c_D \mid \dim \tau$, since $\dim \tau = c_D \dim \rho$ for its unique preimage ρ. In the example $(r, n, F) = (2, 3, \mathbb{Q}_3(\zeta_3))$, $c_D = 27$, while the covered-torus argument of Proposition 8.1 yields only the weaker factor $b_D = 3$ in the dimension constraints. A genuine irreducible of dimension not divisible by 27 would disprove *integrality of this specific transform* L_D, but would not disprove the existence of some other wild-place correspondence. The existence of such an irreducible is an open computational test, not an asserted example.

Remark 5.6 (Why unitarity is not integrality)**.** A unitary operator need not send an orthonormal basis to itself. Even if its matrix is locally finite, the entries need not be integral. A trace-formula or representation-theoretic positivity argument is needed to show $a_{\rho\tau} \in \mathbb{Z}$. The remaining genuinely unresolved local step is the integrality statement at *wild places*, where $p \mid n$; see Section 8. It should not be concealed under the phrase "unitary character transfer."

6 Adams operations and finite-group characters

We recall the K-theoretic origin of the Adams operations and explain why they preserve irreducibility in certain tame finite-group situations.

Background from K-theory and finite-group characters

The Adams operations (after J. F. Adams) originated in algebraic topology, where complex topological K-theory studies the Grothendieck group of complex vector bundles. For a finite group Q, a Q-equivariant complex vector bundle over a point is simply a finite-dimensional complex representation of Q. Consequently

$$K_Q^0(\mathrm{pt}) \cong R(Q),$$

the representation (or character) ring [5]. The operations $\lambda^j[V] = [\bigwedge^j V]$ give this ring its λ-ring structure. The Adams operations ψ^k are the corresponding Newton-polynomial operations; for instance

$$\psi^1(V) = V, \qquad \psi^2(V) = V \otimes V - 2\bigwedge^2 V \quad \text{in } R(Q).$$

This description connects Adams's topological construction [3] with Atiyah's systematic account of power operations [4].

To see the character-theoretic formula, restrict V to the cyclic subgroup generated by $g \in Q$. On this subgroup V decomposes as a sum of eigenlines with eigenvalues α_i; the defining property $\psi^k(L) = L^{\otimes k}$ on line representations gives

$$\boxed{\chi_{\psi^k(V)}(g) = \sum_i \alpha_i^k = \chi_V(g^k).} \tag{11}$$

The operation is integral on the Grothendieck ring, but need not preserve the positive cone of actual characters. For example, if V is the standard two-dimensional irreducible representation of S_3, then $\psi^2(V) = \mathbf{1} - \operatorname{sgn} + V$, a virtual character. If $\gcd(k, |Q|) = 1$, however, raising the root-of-unity eigenvalues to the kth power is a cyclotomic Galois automorphism; it permutes irreducible characters. Section 7 exploits a more delicate version of this fact with a *fixed central character*, so that a central n-primary subgroup can be retained even when the full finite quotient has order divisible by n. This is why the Adams argument establishes integrality at tame places but not at wild ones.

For a genuine finite-dimensional representation τ of $\widetilde{H}$, let

$$\Psi_n(\Theta_\tau)(h) = \Theta_\tau(h^*) = \operatorname{tr} \tau(\widetilde{h}^n).$$

The nth Adams operation in the complex representation ring of $\widetilde{H}$ is a virtual character with integer coefficients. Since μ_n acts on τ by scalars, its nth Adams operation is trivial on μ_n and descends to a virtual character of H. Thus Ψ_n is integral and has central character θ.

Proposition 6.1 ([**P**] Adams pullback and its weight)**.** *For $\tau \in \operatorname{Irr}_{\widetilde{\theta}}(\widetilde{H})$, the class function $\Psi_n(\Theta_\tau)$ is a virtual character in the integral representation ring $R_\theta(H)$. On regular $t \in H$,*

$$\Psi_n(\Theta_\tau)(t) = |n|_F^{-(r-1)/2}\, R(t)\, (L_D^{-1}\Theta_\tau)(t). \tag{12}$$

Proof. After a suitable unramified twist, a given finite-dimensional smooth representation has finite image, and hence factors through a finite quotient Q of $\widetilde{H}$. The usual nth Adams operation is a virtual character of Q. For $\zeta \in \mu_n$ it satisfies $\psi^n\Theta(\zeta g) = \Theta((\zeta g)^n) = \Theta(g^n)$, so linear independence of irreducible characters shows that all its constituents have trivial μ_n-action. Thus it descends to H, with the required central character. Undoing the unramified twist preserves this assertion (it changes under the nth power in the expected way). The displayed formula is the inverse of (6) at $z = 1$. □

In general the nonconstant weight $R(t)^{-1}$ prevents us from deducing integrality of L_D from integrality of the Adams operation alone. In Section 7, however, we prove $R \equiv 1$ when $p \nmid n$, and then obtain a complete tame-case character correspondence. At wild places, the discriminant weight is generally nontrivial.

7 The tame case: splitting and irreducible character integrality

In this section assume that the residue characteristic p does not divide n. We use the special cover of Part I, including the centrality of $\widetilde{Z}$ and its good-element calculation. Write $H^1 = \ker(\mathrm{Nrd} : H \to F^\times)$, and put $U_D^1 = 1 + \mathfrak{P}_D$ for the principal units in the maximal order of D. The term *tame* here means $p \nmid n$; it does not impose a condition on the ramification of a maximal subfield of D.

Lemma 7.1 ([**P**] Prime-to-n finite quotients)**.** *All finite continuous quotients of H^1 and H/Z have order relatively prime to n. In particular both groups have continuous cohomology $H^i(H^1, \mu_n) = 0$ for $i = 1, 2$, where H^1 acts trivially on μ_n.*

Proof. Write $q = |\mathcal{O}_F/\mathfrak{p}_F|$. The standard residue-field quotient of the norm-one group embeds in

$$\ker\big(N_{\mathbb{F}_{q^r}/\mathbb{F}_q} : \mathbb{F}_{q^r}^\times \longrightarrow \mathbb{F}_q^\times\big), \qquad M := \frac{q^r - 1}{q - 1};$$

the kernel of reduction is pro-p. In fact the displayed finite group is the full residue quotient, but divisibility of its order by M is all that is needed. Since $\mu_n \subset F$ and $p \nmid n$, $q \equiv 1 \pmod{n}$, and therefore $M \equiv r \pmod{n}$. Consequently $\gcd(M, n) = 1$ by $\gcd(r, n) = 1$, and no finite quotient of H^1 can have order divisible by a prime factor of n.

For H/Z, the image of U_D^1 is pro-p, and the residual quotient has an extension with one factor of order dividing M and another of order r (the valuation modulo scalar valuations). Thus every finite quotient of H/Z has order prime to n as well. For a profinite group all of whose finite quotients have order prime to n, continuous cohomology with trivial finite μ_n-coefficients vanishes in positive degrees: every cocycle factors through a finite quotient, where averaging is available. □

Proposition 7.2 ([**P**] Unique splitting over the norm-one group)**.** *The restriction of the cover to H^1 admits a unique continuous homomorphic splitting $s : H^1 \to \widetilde{H}$. Its image is normal in $\widetilde{H}$. The quotient*

$$\mathcal{A} = \widetilde{H}/s(H^1)$$

is an abelian locally compact group, fitting into a central extension $1 \to \mu_n \to \mathcal{A} \to F^\times \to 1$. For each prescribed faithful character ϵ of μ_n there exists some *continuous unitary character $\nu : \widetilde{H} \to \mathbb{C}^\times$ with $\nu|_{\mu_n} = \epsilon$ and $\nu|_{s(H^1)} = 1$.*

Proof. Vanishing of $H^2(H^1, \mu_n)$ in Lemma 7.1 provides a continuous splitting, while vanishing of H^1 gives uniqueness. Conjugating a splitting by a lift of an element of H produces another splitting of the normal group H^1; uniqueness therefore proves normality.

The reduced norm identifies H/H^1 with $F^\times$. The commutator of two elements of $\mathcal{A}$ gives a continuous alternating bicharacter $c : F^\times \times F^\times \to \mu_n$. The preimage of scalar elements is central in $\widetilde{H}$, and the reduced norm of a scalar a is a^r. Hence $c(a^r, b) = 1$. By bilinearity $c(a, b)^r = 1$, while $c(a, b)^n = 1$ since its image lies in μ_n. Coprimality of r, n implies $c = 1$, so $\mathcal{A}$ is abelian. Every continuous unitary character of a closed subgroup of a locally compact abelian group extends to such a character of the whole group. Apply this to the finite closed subgroup $\mu_n \subset \mathcal{A}$ and compose with $\widetilde{H} \to \mathcal{A}$. □

Remark 7.3 (Twisting is not the character transfer)**.** A choice of ν produces an elementary dimension-preserving bijection between irreducible ordinary representations and genuine irreducible representations, via $\rho \mapsto \nu \otimes (\rho \circ p)$. This statement concerns the *full* genuine dual, with central characters adjusted by ν. It is not, by itself, the assertion that the canonical transform L_D realizes this particular bijection. Moreover a prescribed $\widetilde{\theta}$ need not equal $\nu|_{\widetilde{Z}}$; keeping track of this difference is necessary.

Lemma 7.4 ([**P**] Tame discriminant identity)**.** *For every $h \in H \setminus Z$ one has*

$$\boxed{\Delta(h^n) = \Delta(h), \qquad R(h) = 1.}$$

The locally constant extension of R to the scalar center is also identically one.

Proof. The compact profinite group $K = H/Z$ has all finite quotients of order prime to n by Lemma 7.1. On each finite quotient Q the power map $x \mapsto x^n$ is bijective (its inverse is $x \mapsto x^m$ for any m with $mn \equiv 1 \pmod{\exp Q}$). Consequently $P_n : K \to K$ is a homeomorphism which preserves probability Haar measure: it permutes the cosets of every open normal subgroup. It is an analytic local diffeomorphism, since its derivative is invertible at regular points by Lemma 3.1, and at scalars it is multiplication by $n \neq 0$ on the Lie algebra.

At a regular element h, write $h\exp(X)$ for a local left-trivialized chart. The derivative of the power map on $\mathfrak{h}/\mathfrak{z}$ is

$$(dP_n)_h = \sum_{k=0}^{n-1} \mathrm{Ad}(h)^{-k}.$$

On the $r-1$ dimensional subspace $\mathfrak{t}/\mathfrak{z}$, $\mathrm{Ad}(h)$ acts trivially, so these directions contribute $|n|_F^{r-1}$ to the absolute determinant. Over a splitting field, the complementary root directions have eigenvalues $u_{ij} = \lambda_i/\lambda_j$ for $i \neq j$ under $\mathrm{Ad}(h)$. The corresponding eigenvalues of the derivative are

$$\sum_{k=0}^{n-1} u_{ij}^{-k} = \frac{1 - u_{ij}^{-n}}{1 - u_{ij}^{-1}},$$

which are nonzero by Lemma 3.1. Taking the product over $i \neq j$ (equivalently, the F-determinant) yields

$$|\det(dP_n)_h|_F = |n|_F^{r-1} \prod_{i \neq j} \left| \frac{1 - u_{ij}^{-n}}{1 - u_{ij}^{-1}} \right|_F = |n|_F^{r-1} \frac{\Delta(h^n)^2}{\Delta(h)^2}.$$

Here the middle product is interpreted with the appropriate extension of the normalized absolute value to a splitting field. The p-adic analytic change-of-variables theorem [11, Proposition 1.2.1] applies to the local analytic diffeomorphism P_n. The probability Haar measure is associated locally to a nonvanishing left-invariant top-degree differential form, so preservation of Haar measure implies $|\det(dP_n)_h|_F = 1$ almost everywhere. The absolute determinant is locally constant and nonzero on the regular set, hence is 1 at every regular h. Since $|n|_F = 1$, the discriminant identity follows. The value at scalars is 1 by Lemma 3.3. □

Lemma 7.5 ([**P**] Adams operations with fixed central character)**.** *Take $G = D^\times$ and $Z_G = F^\times$, and assume $p \nmid n$. For every smooth unitary character δ of Z_G, the nth Adams operation gives a bijection*

$$\psi^n : \mathrm{Irr}_\delta(G) \xrightarrow{\sim} \mathrm{Irr}_{\delta^n}(G), \qquad \Theta_{\psi^n\sigma}(g) = \Theta_\sigma(g^n),$$

with central characters interpreted on the same scalar center.

Proof. First consider a finite group Q with central subgroup C and $|Q/C|$ prime to n. Let π be the set of primes dividing n, and let Q_π be the π-primary subgroup of the finite abelian group C. Since $|Q/C|$ is prime to n, we have $|Q : Q_\pi| = |Q/C|\,|C : Q_\pi|$ prime to n. Thus Q_π is a *normal Hall π-subgroup* of Q (rather than a single-prime Sylow subgroup); it is central because $Q_\pi \subset C$. Schur–Zassenhaus provides a complement $Q_{\pi'}$, and centrality gives $Q = Q_\pi \times Q_{\pi'}$, with $|Q_{\pi'}|$ prime to n. Fixing the central character on C fixes the one-dimensional representation of Q_π. On $Q_{\pi'}$ the

operation ψ^n is the cyclotomic Galois automorphism $\zeta \mapsto \zeta^n$ of character values; since n is prime to $|Q_{\pi'}|$, it permutes irreducible characters. On Q_π it raises the fixed central character to its nth power. Therefore ψ^n bijects irreducibles of central character δ with those of central character δ^n. This is a statement *at a fixed* δ: the operation need not be injective on a union of different central-character fibres, since distinct characters whose quotient has order dividing n can have the same nth power.

For $G = D^\times$, given δ choose a unitary unramified character $\eta = \xi \circ \mathrm{Nrd}$ of G so that $\delta\eta|_Z$ is trivial on a chosen scalar uniformizer ϖ_F (choose an rth root of the inverse scalar value). Then every smooth irreducible with central character $\delta\eta|_Z$ becomes a smooth representation of the compact profinite group $G/\langle\varpi_F\rangle$; it factors through a finite quotient, and its image central subgroup has quotient of order prime to n. For the reverse direction, choose the open normal subgroup through which a given representation factors small enough that $\delta\eta|_Z$ is trivial on its intersection with the center; this is possible because the center modulo $\langle\varpi_F\rangle$ is a closed subgroup of the compact profinite quotient. Then both the specified central character and its nth power factor through the same finite central subgroup. The preceding finite-group argument applies to these representations. Undo the twist, observing that $\psi^n(\eta \otimes \sigma) = \eta^n \otimes \psi^n\sigma$. The same finite-quotient argument applied in reverse proves surjectivity. Note that this argument keeps the central π-part of each finite quotient: an unramified twist cannot generally remove n-torsion coming from the units of Z. □

Theorem 7.6 ([**P**] Tame irreducible character transfer)**.** *Assume $p \nmid n$ and the covering data of Part I. For each unitary genuine central character $\widetilde{\theta}$ on $\widetilde{Z}$, put $\theta(a) = \widetilde{\theta}(a^*)$. Then L_D maps irreducible characters bijectively onto genuine irreducible characters,*

$$\boxed{L_D : \{\Theta_\rho : \rho \in \mathrm{Irr}_\theta(H)\} \xrightarrow{\sim} \{\Theta_\tau : \tau \in \mathrm{Irr}_{\widetilde{\theta}}(\widetilde{H})\}.}$$

Every representation on the right corresponding to ρ has $\dim\tau = \dim\rho$. In particular the character matrix of Corollary 5.1 is a permutation matrix in the tame case.

Proof. Choose the genuine one-dimensional character ν given by Proposition 7.2, with $\nu|_{\mu_n} = \widetilde{\theta}|_{\mu_n}$. Put

$$\chi_\nu(h) = \nu(\widetilde{h})^n \quad (h \in H), \qquad \delta(a) = \frac{\widetilde{\theta}(\widetilde{a})}{\nu(\widetilde{a})} \quad (a \in F^\times).$$

These are well-defined ordinary characters; in particular χ_ν is defined on all of H, while δ is a character of Z. By the definition of the star map,

$$\theta(a) = \chi_\nu(a)\,\delta(a)^n.$$

For $\rho \in \mathrm{Irr}_\theta(H)$, the representation $\chi_\nu^{-1} \otimes \rho$ has central character δ^n. By Lemma 7.5 there is a unique $\sigma \in \mathrm{Irr}_\delta(H)$ such that

$$\Theta_\rho(t) = \chi_\nu(t)\,\Theta_\sigma(t^n) \qquad (t \in H).$$

Define the genuine irreducible representation $\tau = \nu \otimes (\sigma \circ p)$. It has central character $\widetilde{\theta}$, and for regular t

$$\Theta_\tau(t^*) = \nu(t^*)\,\Theta_\sigma(t^n) = \chi_\nu(t)\,\Theta_\sigma(t^n) = \Theta_\rho(t).$$

By Lemma 3.2 the power map is an automorphism of each compact torus $E^\times/F^\times$ when $p \nmid n$. Thus every regular element of $\widetilde{H}$ is of the form zt^* for some $z \in \widetilde{Z}$ and regular t. The same equality holds on all regular elements by central equivariance. Since $b_D = 1$ and $R = 1$ by Lemma 7.4, this is exactly $L_D\Theta_\rho = \Theta_\tau$. The map is bijective by the corresponding bijection for ψ^n and by tensoring with ν. Finally tensoring with one-dimensional characters and the finite-group Galois operation preserve dimension. □

Remark 7.7 (Independence of the auxiliary genuine character)**.** Although the construction in the proof chooses a genuine one-dimensional character ν, the resulting correspondence is independent of that choice. Indeed, L_D was defined solely from $\widetilde{\theta}$, the star map, and the normalized discriminant. For each ρ, the character $L_D\Theta_\rho$ determines a unique irreducible genuine representation of $\widetilde{H}$; this follows from orthogonality (or linear independence) of irreducible characters. Consequently any permissible choice of ν produces the same isomorphism class τ. This is a *canonical correspondence relative to the fixed covering datum and central characters*, not an assertion that ν itself is canonical.

Compatibility audit preceding Corollary 7.8. We distinguish the intrinsically defined nth power of a lift from any choice of cocycle section. The following facts are the exact compatibilities needed to compose the two local correspondences; they are consequences of Part I and the formulas in [7, §§2.2, 3.1], not additional assumptions.

Write $F = K_v$ and $c_v = r'c$ as in Part I, with $rr' \equiv 1 \pmod{n}$. Part I, Proposition 2.3(b), identifies the restriction of the ambient cover, after the prescribed relabeling of μ_n, with the Kazhdan–Patterson extension whose parameter is c_v. In the standard parameter-zero section the cocycle on the diagonal torus is $\sigma_F^{(0)}(a,b) = \prod_{i<j}(a_i, b_j)_n$; adding the determinant factor gives

$$\sigma_F^{(c_v)}(a,b) = \prod_{i<j}(a_i,b_j)_n \left(\prod_i a_i, \prod_j b_j\right)_n^{c_v}. \tag{13}$$

This is precisely Zou's block formula [7, (2.1)] in the one-dimensional-block case. In particular the two models determine the same extension class and the same centrality parameter

$$d_r = \gcd(n, r - 1 + 2rc_v) = n.$$

Here the claimed identification is of *extensions*; Part I does not claim a preferred equality of two cocycles on all pairs of nondiagonal matrices.

On scalar matrices aI_r, bI_r, equation (13) becomes

$$\sigma_F^{(c_v)}(aI_r, bI_r) = (a,b)_n^{e_r}, \qquad e_r = \binom{r}{2} + c_v r^2, \qquad 2e_r = r(r - 1 + 2rc_v). \tag{14}$$

Since n is odd and divides $r - 1 + 2rc_v$, it divides e_r. Consequently the *standard* Kazhdan–Patterson section is multiplicative on scalar matrices, and for $a \in F^\times$ its nth power satisfies $(s(aI_r))^n = s(a^n I_r)$. In particular if $\zeta \in \mu_n(F)$, then $(\zeta I_r)^* = 1$. Thus our scalar character $\theta(a) = \widetilde{\theta}(a^*)$ satisfies $\theta|_{\mu_n} = 1$ and is exactly Zou's required relation $\widetilde{\omega}(s(a^n I_r)) = \omega(a)$, after transporting $\widetilde{\theta}$ through an isomorphism of covers.

Flicker–Kazhdan define the star map for odd n by $x^* = \widetilde{x}^{\,n}$ [6, §4]; this agrees with (2) and is natural under *every* isomorphism of central extensions fixing μ_n. Two identifications of the ambient GL_r cover with the Kazhdan–Patterson model differ by $\chi \circ \det$ for a character $\chi : F^\times \to \mu_n$ (Part I, Proposition 2.3(c)). Their values agree on star images, since $\chi(\det(x^n)) = 1$. At arbitrary scalar translates one transports the central character through the same identification. Hence this choice does not alter the eventual identity for genuine characters of the original ambient cover.

Finally, Part I, Theorem 3.9(b), identifies the covered elliptic tori by conjugation inside the common ambient cover and is the identity on their common central subgroup. It therefore satisfies

$$\jmath(t_D^*) = (j(t))^*_{\mathrm{GL}}, \qquad \jmath(zt_D^*) = z\,(j(t))^*_{\mathrm{GL}} \quad (z \in \widetilde{Z}). \tag{15}$$

The Weyl discriminants on $D^\times$ and $\mathrm{GL}_r(F)$ are the same expression in the conjugate eigenvalues of t; see (3) and Part I, Definition 4.1. These are character *values*, not global orbital integrals: no comparison of arbitrary quotient Haar measures is being inferred.

Corollary 7.8 ([**P**] Tame character relation in the available Flicker–Kazhdan range)**.** *Let $F = K_v$ be a finite place at which D is division, $p \nmid n$, and the standing odd-n and $(n, r) = 1$ assumptions hold. Fix the common faithful genuine central character $\widetilde{\theta}$ of Part I and let $\theta(a) = \widetilde{\theta}(a^*)$. Assume that the square-integrable Flicker–Kazhdan correspondence for the parameter-c_v Kazhdan–Patterson cover is available in the corrected form of [7, Proposition 3.6 and Definition 3.7]. A sufficient published range for this external input is $(n, N_r) = 1$, where N_r is the least common multiple of the composite positive integers at most r, with $N_r = 1$ when there are none [6, §26].*

Then the division-side tame transfer L_D of Theorem 7.6, the classical local Jacquet–Langlands correspondence, and the corrected metaplectic correspondence compose to a bijection between the relevant genuine irreducible representations $\widetilde{\pi}_D$ of $\widetilde{D^\times}$ and genuine essentially square-integrable representations $\widetilde{\pi}$ of $\widetilde{G}_v$ with the specified central character. On matching regular good elements,

$$\boxed{\Theta_{\widetilde{\pi}_D}(\widetilde{\gamma}) = (-1)^{r-1}\Theta_{\widetilde{\pi}}(\jmath(\widetilde{\gamma})).} \tag{16}$$

Thus relation (D2) of Part I is proved in this stated tame range, conditional only *on the cited external Flicker–Kazhdan theorem and the established local results of Parts I–III; it is not asserted for all (n, r).*

Proof. 1. Matching characters on the division side. By Theorem 7.6 there is a unique $\rho \in \mathrm{Irr}_\theta(D^\times)$ for which $L_D\Theta_\rho = \Theta_{\widetilde{\pi}_D}$. At $p \nmid n$ we have $b_D = 1$ and $\Delta(t^n) = \Delta(t)$ (Lemma 7.4); thus

$$\Theta_{\widetilde{\pi}_D}(zt_D^*) = \widetilde{\theta}(z)\Theta_\rho(t) \qquad (t \in E^\times \setminus F^\times, z \in \widetilde{Z}). \tag{17}$$

Every regular good element has this form (Part I, Proposition 3.5(a)), and the power map is injective on $E^\times/F^\times$ by Lemma 3.2.

2. Classical Jacquet–Langlands. Let $\pi = \mathrm{JL}(\rho)$, with the same ordinary scalar character θ. The usual elliptic-character relation is

$$\Theta_\rho(t) = (-1)^{r-1}\Theta_\pi(j(t)), \tag{18}$$

where $j : E \hookrightarrow M_r(F)$ and the discriminant normalization is the common one above [9]. The character θ is trivial on μ_n by (14), so π lies in the permitted linear central-character family in Zou's correspondence.

3. Explicit evaluation of Zou's root sum. Let $x = j(t)$ be regular elliptic. If $y^n = x^n$ in $\mathrm{GL}_r(F)$, then y commutes with x^n. As t^n is regular (Lemma 3.1), its centralizer is $j(E^\times)$; therefore $y \in j(E^\times)$ and $(yx^{-1})^n = 1$. Since $\mu_n(E) = \mu_n(F)$, the solutions are exactly the n scalar multiples $y = \zeta x$ with $\zeta \in \mu_n$. For every one, $\Delta(y) = \Delta(x)$ and $\Theta_\pi(y) = \Theta_\pi(x)$ because $\theta(\zeta) = 1$. Equation (14) gives $(\zeta I_r)^* = 1$, so the phase $\epsilon(x^*/y^*)$ in Zou's equation (3.2) is also 1. Zou's corrected root-sum formula therefore reduces to

$$\Theta_{\widetilde{\pi}}(x^*) = \frac{n}{d_F(d_r)|n|_F^{r/2}}\frac{\Delta(x)}{\Delta(x^n)}\Theta_\pi(x) = b_D\frac{\Delta(x)}{\Delta(x^n)}\Theta_\pi(x), \tag{19}$$

where $d_F(m) = m|m|_F^{-1/2}$ and $d_r = \gcd(n, r - 1 + 2rc_v) = n$. Indeed,

$$\frac{n}{d_F(n)|n|_F^{r/2}} = \frac{n}{n|n|_F^{-1/2}|n|_F^{r/2}} = |n|_F^{-(r-1)/2} = b_D.$$

The equality of numerical factors is valid even at wild places; it is *not* a wild-place character-integrality theorem. Here $p \nmid n$ implies $b_D = 1$, and Lemma 7.4 gives $\Delta(x) = \Delta(x^n)$. Consequently $\Theta_{\widetilde{\pi}}(x^*) = \Theta_\pi(x)$.

4. Transfer back to the two actual covers. By (15), the star values in the last identity coincide with those obtained from the covered division-algebra torus. Multiplying by $\widetilde{\theta}(z)$ and combining (17), (18) and (19) yields (16) for every good regular element. The star and central-character identifications in the preceding compatibility audit show that the conclusion is independent of the auxiliary isomorphism to the standard Kazhdan–Patterson cocycle model.

5. Bijection rather than just a character identity. Theorem 7.6 is a bijection on genuine irreducibles with fixed central character; classical local Jacquet–Langlands is a bijection onto the corresponding essentially square-integrable linear representations; and Zou's corrected correspondence is a bijection in the specified external range and central-character family. Their composition has an inverse obtained by reversing the three maps. This concludes the claim without inferring irreducibility or integrality from L^2 unitarity alone. □

Remark 7.9 (Scope and status of Corollary 7.8)**.** The earlier version of this corollary carried status **[O]** because the cocycle, star map and central characters had not been compared explicitly with Zou. Equations (13)–(15) now perform that comparison; the result is **[P]** *under the written external correspondence hypothesis.* In particular $(n, N_r) = 1$ suffices by the stated Flicker–Kazhdan theorem, as used by Zou. For additional parameter pairs one must verify an applicable extension of that external input; the present proof does not supply it. At places $p \mid n$, integrality of L_D remains open. Global matching of Haar measures, test-function normalizations, and unramified spherical units is also not proved by this local pointwise character identity.

8 Places dividing n: non-splitting and constraints

At residue characteristics dividing n, the tame splitting argument fails. The commutator pairing from Part I gives a precise obstruction, but does not supply an irreducible-character correspondence. The numerical test of Remark 5.5 is an additional constraint on any proposed correspondence.

Proposition 8.1 (**[P]** Wild-place obstruction to splitting)**.** *Suppose $p \mid n$ and retain the commutator formula and nondegeneracy assertion of Part I, Proposition 3.5(a),(c). Then $\widetilde{H}$ does not split over H^1, and it has no genuine one-dimensional representations. More precisely, every genuine irreducible representation of $\widetilde{H}$ has dimension divisible by*

$$b_D = |n|_F^{-(r-1)/2} > 1.$$

The existence or integrality of the wild-place transform L_D on irreducible characters is not established by this proposition.

Proof. Put $E^1 = \ker(N_{E/F} : E^\times \to F^\times)$ and $B = E^\times/(F^\times E^{\times n})$. The group B has order $|n|_F^{-(r-1)} > 1$ by Lemma 3.2; it is n-torsion. The map $E^1 \to B$ is surjective: for $t \in E^\times$,

$$s_t = t^r/N_{E/F}(t) \in E^1, \qquad [s_t] = [t]^r \in B.$$

Since $\gcd(r, n) = 1$, raising to the rth power is an automorphism of B, so these elements generate B. The commutator pairing on the covered torus, described in Part I, Proposition 3.5(a), has radical $F^\times E^{\times n}$; it is therefore nondegenerate on B. Restricting to E^1 preserves a nontrivial pairing because $E^1 \to B$ is surjective. Consequently the inverse image of $E^1 \subset H^1$ in $\widetilde{H}$ is nonabelian, so the cover cannot admit a splitting over H^1.

Part I, Proposition 3.5(c), identifies the dimensions of genuine irreducible representations of the covered torus with $\sqrt{|B|} = b_D$ (for the fixed genuine central character). Restricting a

finite-dimensional genuine representation of $\widetilde{H}$ to the covered torus and decomposing it into these constituents proves the divisibility assertion, which excludes genuine one-dimensional representations. □

Remark 8.2 (Global use and the remaining difficulty)**.** At a finite place v with $p_v \nmid n$, the local $D_v^\times$-side integrality problem is now resolved under the stated covering assumptions. At places dividing n it remains a distinct and genuinely nonabelian problem. Global division algebras can be chosen to avoid those places subject to the Brauer-invariant sum and local-degree conditions. This removes the *division-side local integrality obstacle* for such a choice, but does not prove global trace-formula comparison, spectral multiplicity statements, or a metaplectic Jacquet–Langlands theorem at the split places.

9 Local orbital-integral transfer on $D^\times$

In this section no assumption $p \nmid n$ is imposed. The construction therefore applies at both tame and wild places; it uses the good-element description of Proposition 2.4. We keep the quotient probability Haar measures from Section 4. The result is a transfer of test functions and geometric distributions; it is not an assertion that the transform of every irreducible character is irreducible.

9.1 Test functions and orbital integrals

For a unitary central character η of Z, let

$$\mathcal{C}_{\eta^{-1}}(H) = \{f : H \to \mathbb{C} : \ f \text{ is locally constant and } f(ah) = \eta(a)^{-1} f(h) \text{ for } a \in Z\}.$$

We use analogous notation for $\mathcal{C}_{\widetilde{\eta}^{-1}}(\widetilde{H})$. Since H/Z and $\widetilde{H}/\widetilde{Z}$ are compact, all these functions are compactly supported modulo the indicated center. These are the test-function central characters *inverse* to the central characters of representations: $\Theta_\rho(ah) = \theta(a)\Theta_\rho(h)$, whereas $f(ah) = \theta(a)^{-1} f(h)$, so that the product is a function on H/Z.

Write dh for Haar probability measure on H/Z, and define the conjugation average

$$(\mathcal{A}_H f)(h) = \int_{H/Z} f(x^{-1}hx)\, dx. \tag{20}$$

For the cover define, using any lift $\widetilde{x}$ of x,

$$(\mathcal{A}_{\widetilde{H}} \widetilde{f})(\widetilde{h}) = \int_{H/Z} \widetilde{f}(\widetilde{x}^{-1}\widetilde{h}\widetilde{x})\, dx. \tag{21}$$

Centrality of μ_n and $\widetilde{Z}$ makes the latter integrand independent of the chosen lifts and of the representative of $x \in H/Z$. Both averages are smooth, conjugation invariant, and central equivariant; they satisfy $\mathcal{A}_H^2 = \mathcal{A}_H$ and $\mathcal{A}_{\widetilde{H}}^2 = \mathcal{A}_{\widetilde{H}}$.

For $t \in T^{\mathrm{reg}}$, $T = F(t)^\times$, equip H/T with the quotient *probability* measure induced by those on H/Z and T/Z. Define

$$O_H(t, f) = \int_{H/T} f(x^{-1}tx)\, d\dot{x} = (\mathcal{A}_H f)(t). \tag{22}$$

For a good regular $\widetilde{t} \in p^{-1}(T)$ the corresponding genuine orbital integral is

$$O_{\widetilde{H}}(\widetilde{t}, \widetilde{f}) = \int_{H/T} \widetilde{f}(\widetilde{x}^{-1}\widetilde{t}\widetilde{x})\, d\dot{x} = (\mathcal{A}_{\widetilde{H}} \widetilde{f})(\widetilde{t}). \tag{23}$$

The first integral is independent of coset representatives precisely because $\widetilde{t}$ is good. If $\widetilde{t}$ is not good, the full conjugation average in (21) vanishes: replacing $\widetilde{x}$ by $\widetilde{u}\widetilde{x}$, for a lift of an element of T with nontrivial commutator against $\widetilde{t}$, multiplies the averaged value by a nontrivial genuine scalar. We use this vanishing rather than attempting to define an H/T integral for a nongood element.

For later use put

$$J_H(t,f) = \Delta(t) O_H(t,f), \qquad J_{\widetilde{H}}(\widetilde{t},\widetilde{f}) = \Delta(p(\widetilde{t})) O_{\widetilde{H}}(\widetilde{t},\widetilde{f}) \tag{24}$$

when $\widetilde{t}$ is good. The factor Δ is the *square root* of the ordinary Weyl discriminant as in (3).

9.2 An explicit transfer theorem

Let $L_D^{(\widetilde{\eta})}$ denote the map in Definition 4.1 for the genuine central character $\widetilde{\eta}$, with its associated ordinary central character $\eta(a) = \widetilde{\eta}(a^*)$. In particular, $L_D^{(\widetilde{\theta}^{-1})}$ acts on invariant test functions having the inverse central character.

Theorem 9.1 ([**P**] Division-side test-function transfer)**.** *Assume the specified cover of Part I and Proposition 2.4. Fix a genuine central character $\widetilde{\theta}$ and put $\theta(a) = \widetilde{\theta}(a^*)$. The following maps are well defined:*

$$\mathcal{O}_{D,\widetilde{\theta}} : \mathcal{C}_{\theta^{-1}}(H) \longrightarrow \mathcal{C}_{\widetilde{\theta}^{-1}}(\widetilde{H}), \qquad \mathcal{O}_{D,\widetilde{\theta}} f = L_D^{(\widetilde{\theta}^{-1})}(\mathcal{A}_H f), \tag{25}$$

$$\mathcal{B}_{\widetilde{\theta}} : \mathcal{C}_{\widetilde{\theta}^{-1}}(\widetilde{H}) \longrightarrow \mathcal{C}_{\theta^{-1}}(H), \qquad \mathcal{B}_{\widetilde{\theta}} \widetilde{f} = (L_D^{(\widetilde{\theta}^{-1})})^{-1}(\mathcal{A}_{\widetilde{H}} \widetilde{f}). \tag{26}$$

Their images consist of conjugation-invariant test functions; and

$$\mathcal{B}_{\widetilde{\theta}} \mathcal{O}_{D,\widetilde{\theta}} = \mathcal{A}_H, \qquad \mathcal{O}_{D,\widetilde{\theta}} \mathcal{B}_{\widetilde{\theta}} = \mathcal{A}_{\widetilde{H}}.$$

They therefore induce mutually inverse linear bijections on the respective test-function spaces modulo the kernels of their conjugation-averaging (equivalently, normalized orbital-integral) maps.

More explicitly, for every regular $t \in T$ and $z \in \widetilde{Z}$,

$$\boxed{O_{\widetilde{H}}(z\,t^*, \mathcal{O}_{D,\widetilde{\theta}} f) = b_D\, \widetilde{\theta}(z)^{-1} \frac{\Delta(t)}{\Delta(t^n)} O_H(t,f), \qquad b_D = |n|_F^{-(r-1)/2}.} \tag{27}$$

Consequently the normalized orbital integrals satisfy

$$\boxed{J_{\widetilde{H}}(z\,t^*, \mathcal{O}_{D,\widetilde{\theta}} f) = b_D\, \widetilde{\theta}(z)^{-1} J_H(t,f).} \tag{28}$$

At nongood regular lifts the genuine orbital-integral distribution vanishes. Conversely every genuine test function $\widetilde{f}$ has the same regular orbital integrals as $\mathcal{O}_{D,\widetilde{\theta}}(\mathcal{B}_{\widetilde{\theta}} \widetilde{f})$.

Proof. Conjugation averaging is a projection onto the invariant locally constant functions, and (22)–(23) identify the averages with the stated orbital integrals. The local smoothness and invertibility of L_D on invariant test functions follow from Lemma 4.2 and the proof of Theorem 4.3: on each regular torus quotient the star map is a homeomorphism from T/Z onto the open subgroup $(T/Z)^n$, and near scalar elements it is a local analytic isomorphism. Thus both displayed maps take smooth test functions to smooth test functions. Since the image of $\mathcal{A}_H$ consists of invariant functions, $\mathcal{A}_{\widetilde{H}} \mathcal{O}_{D,\widetilde{\theta}} f = \mathcal{O}_{D,\widetilde{\theta}} f$; applying the inverse L_D gives $\mathcal{B}_{\widetilde{\theta}} \mathcal{O}_{D,\widetilde{\theta}} f = \mathcal{A}_H f$. The other composition is identical. At a good point zt^*, the definition of L_D applied to $\mathcal{A}_H f$ gives (27). Multiplying it by $\Delta(t^n)$ gives (28). Remark 2.3 proves the vanishing on nongood lifts. The inverse composition identity gives the converse statement. No assertion about irreducible-character integrality is used. □

Remark 9.2 (Comparison with the root transport of Part II)**.** The orbital-integral operator $\mathcal{O}_{D,\widetilde{\theta}}$ is *not* the local root-section transport denoted $\mathcal{T}$ in Part II. They have distinct domains, definitions and central character conventions. Near the center, the parameters satisfy $t^* = \sigma(t^n)$, with σ the root section in Part II; this identity does not equate the operators on arbitrary test functions. The orbital transfer here also involves conjugation averaging and the factor $b_D\Delta(t)/\Delta(t^n)$.

9.3 The duality with character transfer

The inverse central characters in Theorem 9.1 are forced by trace pairings, not a cosmetic convention.

Proposition 9.3 ([**P**] Trace-pairing adjunction)**.** *For every smooth invariant class function Φ on H of central character θ, and every $f \in \mathcal{C}_{\theta^{-1}}(H)$,*

$$\boxed{\int_{\widetilde{H}/\widetilde{Z}} (L_D^{(\widetilde{\theta})}\Phi)(\widetilde{h})(\mathcal{O}_{D,\widetilde{\theta}}f)(\widetilde{h})\, d\widetilde{h} = \int_{H/Z} \Phi(h)f(h)\, dh.} \tag{29}$$

In particular, for $\rho \in \mathrm{Irr}_\theta(H)$, writing the finite expansion of Corollary 5.1 as $L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau$, one has

$$\sum_\tau a_{\rho\tau} \operatorname{tr}\tau(\mathcal{O}_{D,\widetilde{\theta}}f) = \operatorname{tr}\rho(f). \tag{30}$$

If $p \nmid n$, Theorem 7.6 reduces this equality to $\operatorname{tr}\tau_\rho(\mathcal{O}_{D,\widetilde{\theta}}f) = \operatorname{tr}\rho(f)$ for the corresponding genuine irreducible representation τ_ρ. For $p \mid n$, formula (30) is a distributional identity with possibly nonintegral coefficients, not *a representation correspondence.*

Proof. Because Φ is invariant, averaging f does not change the integral on the right. For the left side note that $L_D^{(\widetilde{\theta}^{-1})}(\mathcal{A}_H f)$ is the complex conjugate of $L_D^{(\widetilde{\theta})}(\overline{\mathcal{A}_H f})$: the coefficient $b_D\Delta(t)/\Delta(t^n)$ is positive real, and complex conjugation interchanges the two central characters. The unitarity identity of Theorem 4.3, applied to Φ and $\overline{\mathcal{A}_H f}$, proves (29). Equivalently, the Weyl integration formula gives the identity directly: on each T/Z, the product of the two transforms contributes $b_D^2\Delta(t)^2/\Delta(t^n)^2$ and the change of variables (8) contributes I^{-1}; since $b_D^2 = I$, the Weyl integrals are equal. Both trace identities follow by integrating characters against test functions with inverse central character. In particular (30) is a reformulation of unitarity and surjectivity of $\mathcal{O}_{D,\widetilde{\theta}}$, not an independent constraint forcing integrality of the matrix $(a_{\rho\tau})$. □

Corollary 9.4 ([**P**] The identity-element factor)**.** *With the probability measures and central-character conventions of this section, the transferred invariant representative satisfies*

$$\boxed{(\mathcal{O}_{D,\widetilde{\theta}}f)(1) = |n|_F^{-(r^2-1)/2}\, f(1).} \tag{31}$$

The factor equals 1 *at tame places. This is a local pointwise test-function normalization; it is* not *a calculation of the global identity term in a trace formula with different Haar measures and central quotients.*

Proof. The central element 1 is fixed by conjugation, so $(\mathcal{A}_H f)(1) = f(1)$. Evaluate the continuous extension of the transfer at 1, using Lemma 3.3 and $b_D = |n|_F^{-(r-1)/2}$. □

Remark 9.5 (The running wild-place example)**.** For $r = 2$, $n = 3$ and $F = \mathbb{Q}_3(\zeta_3)$, $|3|_F = 1/9$ and $b_D = 3$. Thus $J_{\widetilde{H}}(zt^*, \mathcal{O}_{D,\widetilde{\theta}}f) = 3\widetilde{\theta}(z)^{-1}J_H(t,f)$, whereas $(\mathcal{O}_{D,\widetilde{\theta}}f)(1) = 27f(1)$. This realizes the two normalizations discussed in Remark 5.5.

Remark 9.6 (Scope of local transfer)**.** The existence of such a transfer is elementary on a group compact modulo its center: conjugation averaging identifies orbital integrals with invariant function values. The content of the theorem is the *specified normalization*, extension across singular central elements, and the trace-pairing relation. The local coefficient b_D does not determine adelic orbital measures. The distinction from Part II's root transport is made precise in Remark 9.2.

10 Kottwitz's framework for rational orbits and inner forms

Kottwitz's *Stable trace formula: cuspidal tempered terms* [1] explains how Galois cohomology organizes rational conjugacy classes within stable classes (§§7.7–7.9), how maximal tori are transferred to inner forms (§9.5), and how Tamagawa numbers enter stable trace-formula coefficients (§§5,8). These are statements about *linear* reductive groups; they must not be taken as a metaplectic trace formula. In particular, the stabilization of cuspidal tempered terms sketched in his §12 is conditional, whereas the cohomological calculations used below are independent of that sketch.

Stable and rational classes. For a regular semisimple element $\gamma \in G(K)$ with torus centralizer $T = G_\gamma$, the rational conjugacy classes within its represented stable class are parametrized by

$$\ker\Big[H^1(K,T) \longrightarrow H^1(K,G)\Big] \tag{32}$$

[1, §7.7]. In the present setting, $G = \mathrm{GL}_r$ or $G' = \mathrm{GL}_1(D)$ and $T = \mathrm{Res}_{E/K}\,\mathbb{G}_m$ for a maximal subfield E/K of degree r. By Shapiro's lemma and Hilbert 90, $H^1(K,T) = H^1(E,\mathbb{G}_m) = 1$. Thus every *represented* regular stable class comprises exactly one rational conjugacy class. This does not decide whether the class has a good lift in a covering group.

The inner-form embedding criterion. A degree-r extension E/K embeds in D as a maximal subfield if and only if

$$[D \otimes_K E] = 0 \quad \text{in } \mathrm{Br}(E). \tag{33}$$

By Brauer–Hasse–Noether, this condition can be checked at all completions. Where D_v is division of degree r, the local algebra $E \otimes_K K_v$ must be a field of degree r. Consequently, for these inner forms, local embeddability at every place implies global embeddability. Kottwitz's general §9.5 formulates an obstruction to torus transfer in a class of $H^2(K, T_{\mathrm{sc}})$ (pp. 638–639); the specialized Brauer criterion avoids that additional complication here. For linear orbital-integral transfer see also [10].

Tamagawa factors. Kottwitz §5 identifies the relative Tamagawa number $\tau_1(G) = \tau(G)/\tau(G_{\mathrm{sc}})$ as

$$\tau_1(G) = \frac{|\pi_0(Z(\widehat{G})^{\Gamma_K})|}{|\ker^1(K, Z(\widehat{G}))|}. \tag{34}$$

For GL_r and $\mathrm{GL}_1(D)$ the dual center is $\mathbb{C}^\times$ with trivial Galois action, so this expression is 1. Kottwitz §8 uses relative Tamagawa numbers in stable trace-formula coefficients. Our local b_D instead uses *probability* Haar measures on compact quotients; its conversion to a coherent adelic convention remains a separate calculation, not a consequence of (34).

11 Global power classes and a Hasse principle

The scalar-modulo-nth-power quotient is typically infinite, but its subgroup of classes trivial outside a finite set of places vanishes under the hypotheses below. The proof separates the norm argument from the Kummer–Chebotarev detection step. The following large quotient measures the failure of *global surjectivity* of a power map, but its size is not the correct measure of failure of a *local-to-global principle*. Let E/K be a maximal subfield of D, of degree r, and consider the multiplicative quotient

$$\mathcal{Q}(E/K, n) = E^\times/(K^\times E^{\times n}). \tag{35}$$

A regular rational class represented by $\gamma \in E^\times$ can lie in the scalar multiple of the rational nth-power image only if its class in (35) is trivial. This obstruction must not be confused with the finite local index of Lemma 3.2.

Proposition 11.1 ([**P**] The ambient power quotient is infinite)**.** *Let E/K be a number-field extension of degree at least two and let $n > 1$. Then $\mathcal{Q}(E/K, n)$ is infinite. More precisely, for any $m \geq 1$ and any distinct finite places $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ of K splitting completely in E, there is a surjective homomorphism*

$$\mathcal{Q}(E/K, n) \longrightarrow (\mathbb{Z}/n\mathbb{Z})^m.$$

In particular, even when $\mu_n \subset K$ and $\gcd(n, r) = 1$, the map $[t] \mapsto [t^n]$ on $E^\times/K^\times$ cannot be surjective.

Proof. For each i choose two distinct primes $\mathfrak{P}_{i,1}$ and $\mathfrak{P}_{i,2}$ of E above $\mathfrak{p}_i$. Because $\mathfrak{p}_i$ splits completely, for $a \in K^\times$ the normalized discrete valuations satisfy $v_{\mathfrak{P}_{i,1}}(a) = v_{\mathfrak{P}_{i,2}}(a)$. Consequently

$$\partial_i([x]) = v_{\mathfrak{P}_{i,1}}(x) - v_{\mathfrak{P}_{i,2}}(x) \pmod{n}$$

is well defined on $\mathcal{Q}(E/K, n)$: multiplication by $K^\times$ changes neither difference, and multiplication by an nth power changes it by a multiple of n. The weak approximation theorem for E permits independent prescription of the valuations at this finite collection of primes, so the product $(\partial_1, \ldots, \partial_m)$ is surjective. Chebotarev's theorem supplies infinitely many completely split primes; thus m is arbitrary and the quotient is infinite. □

Remark 11.2 (Effect on the proposed trace formula)**.** The infinitude of $\mathcal{Q}(E/K, n)$ does *not* show that there is an infinite global obstruction among the classes satisfying all local power conditions. Indeed, the classes exhibited by valuation differences are already nontrivial in the local power quotient at the corresponding completely split places. For the split-cover commutator these may also be nongood; that additional implication must be checked in the particular Flicker–Kazhdan cocycle, not inferred merely from the valuation calculation. One must distinguish a large ambient quotient from the kernel of the localization map. That kernel is controlled by the next proposition.

Proposition 11.3 ([**P**] Local-to-global powers modulo scalars)**.** *Let E/K be a separable extension of number fields of degree r, let $n > 1$, and assume $\gcd(n, r) = 1$ and $\mu_n \subset K$. For* every finite set of places Σ *of K, the natural map*

$$\mathcal{Q}(E/K, n) \longrightarrow \prod_{v \notin \Sigma} \frac{(E \otimes_K K_v)^\times}{K_v^\times (E \otimes_K K_v)^{\times n}} \tag{36}$$

is injective. Thus being a scalar times an nth power at almost all places *already implies being so globally.*

Proof. Let $t \in E^\times$ map to the trivial class outside a finite set Σ. Choose integers u, k satisfying $ru + nk = 1$, and set $a = N_{E/K}(t)^u \in K^\times$. For each $v \notin \Sigma$, the assumption supplies $t = a_v y_v^n$ in $(E \otimes_K K_v)^\times$, for $a_v \in K_v^\times$ and $y_v \in (E \otimes_K K_v)^\times$. Taking the norm from this degree-r *étale algebra* produces $N(t) = a_v^r N(y_v)^n$. Consequently

$$\frac{t}{a} = \left(a_v^k \frac{y_v}{N(y_v)^u}\right)^n \quad \text{in } (E \otimes_K K_v)^\times.$$

Thus t/a is an nth power at each completion E_w lying over a place $v \notin \Sigma$. Since $\mu_n \subset E$, a nontrivial Kummer extension $E((t/a)^{1/n})/E$ would be a nontrivial cyclic Galois extension and, by Chebotarev, would possess infinitely many unramified places with nontrivial Frobenius. We may choose one lying over a place $v \notin \Sigma$. At that completion t/a could not be an nth power, a contradiction. Hence $t/a \in E^{\times n}$ and $[t] = 1$ in $\mathcal{Q}(E/K, n)$. □

Corollary 11.4 ([**P**] Vanishing of the associated Sha^1)**.** *Under the assumptions of Proposition 11.3, put $T = \mathrm{Res}_{E/K}\mathbb{G}_m/\mathbb{G}_m$ and let $T[n]$ be its finite n-torsion group scheme. Then for every finite Σ, the restricted localization kernel*

$$\mathrm{Sha}^1_\Sigma(K, T[n]) := \ker\big(H^1(K, T[n]) \to \prod_{v \notin \Sigma} H^1(K_v, T[n])\big)$$

vanishes. In particular, the usual

$$\mathrm{Sha}^1(K, T[n]) := \ker\left[H^1(K, T[n]) \longrightarrow \prod_v H^1(K_v, T[n])\right] = 0. \tag{37}$$

This is stronger than the general finiteness of Sha^1 *for a finite Galois module [2, Chapter I, §4].*

Proof. From $1 \to \mathbb{G}_m \to \mathrm{Res}_{E/K}\mathbb{G}_m \to T \to 1$, Hilbert 90 gives $T(K) = E^\times/K^\times$. The same cohomology sequence gives

$$H^1(K, T) \cong \ker(\mathrm{Br}(K) \to \mathrm{Br}(E)).$$

Restriction followed by corestriction is multiplication by r, so this group is killed by r. Hence, as $\gcd(n, r) = 1$, $H^1(K, T)[n] = 0$. The Kummer exact sequence for multiplication by n on T therefore gives

$$H^1(K, T[n]) \cong T(K)/nT(K) = \mathcal{Q}(E/K, n).$$

At each K_v the same argument applies to the finite étale algebra $E \otimes_K K_v$ (its total degree is still r), yielding analogous isomorphisms for $H^1(K_v, T[n])$. Under these identifications the localization map is (36), which is injective, even after omitting any finite set of places, by the proposition. □

Remark 11.5 (Scope of the Hasse principle)**.** The conclusion uses local power conditions at almost all places, not merely at the division places S. Moreover, an nth-power condition in the scalar quotient must not be identified with the good-element condition for the cover of a split group without a separate commutator calculation. The theorem eliminates this particular Kummer obstruction; it does not provide the missing metaplectic geometric comparison.

12 The global comparison: remaining problems

The following spectral identity is a target rather than a theorem. The arithmetic in Section 11 simplifies its rational-orbit accounting, but leaves the covering-group good-locus and measure

comparisons untouched. Let K be a number field containing μ_n (for odd $n > 1$ this forces K to be totally imaginary, as in Parts I and II), and D/K a central division algebra of prime degree r. Denote by S its finite ramification set. We consider the automorphic spectra of $D^\times(\mathbb{A}_K)$ and its chosen metaplectic cover, both modulo their respective centers. The quotient of $D^\times(\mathbb{A}_K)$ by its rational points and adelic center is compact, hence both spectral trace formulas are discrete. *This fact alone does not establish equality of their geometric sides.*

The following proposed identity is the main mathematical target:

$$\sum_{\widetilde{\Pi}_D} m(\widetilde{\Pi}_D) \operatorname{tr} \widetilde{\Pi}_D(\widetilde{f}) \stackrel{?}{=} \sum_{\Pi_D} m(\Pi_D) \operatorname{tr} \Pi_D(f), \tag{38}$$

for appropriately matched factorizable test functions and compatible global central characters. It is *conditional* on a comparison of the geometric sides with all constants correct. Section 9 records a normalized division-side local matching operator on test functions, dual to the character-space transform. Its existence is automatic from compactness modulo the center, while smoothness and the normalization require the stated local input. Kottwitz's framework in Section 10 clarifies the ordinary rational-orbit and measure issues, but neither automatically gives a geometric comparison for the cover nor settles wild-place character integrality.

12.1 Four proof obligations

(1) **Rational orbit accounting.** Combine the uniqueness of represented regular rational classes from Section 10 with the local-to-global power result of Proposition 11.3. Then determine which rational elements are locally *good* for the particular cover at split and division places. Vanishing of the Kummer localization kernel alone does not identify the surviving cover orbits.

(2) **Local normalization and compatibility.** At $v \in S$, convert the probability-measure test transfer of Section 9 to the fixed global measures and check the local geometric factor against the exact desired global comparison. At split places, cite the exact usable form of Flicker–Kazhdan [6], taking into account Zou's correction to their Lemma 27.1 [7]. Check unramified unit elements and archimedean factors separately.

(3) **Identity and volume terms.** With the trace-formula quotient measures fixed, calculate the identity terms, orbital-integral Jacobians, and central character factors. The b_D of Theorem 4.3 is a *local probability-measure normalization* and does not dispose of the global normalization problem.

(4) **Spectral separation and integrality.** Derive a finite, or otherwise rigorously summable, character identity. Use classical strong multiplicity one for the linear group [8], but do not infer integrality merely from L^2 unitarity. The proposed Cauchy–Schwarz argument requires an explicit identity, its correct normalization, and control of every other local factor.

Remark 12.1 (Restriction on n)**.** The standard hypotheses under which one invokes the original Flicker–Kazhdan trace formula include $(n, N_r) = 1$, where N_r is the least common multiple of the composite integers $\leq r$. Our local Theorem 4.3 uses only n odd, $(n, r) = 1$, and the covering result in Proposition 2.4; it does not use $(n, N_r) = 1$. The global comparison will provisionally impose the latter condition until the exact later improvements or alternatives are verified.

12.2 Conditional representation-theoretic consequences

If (38) is established in a form that proves character integrality at the remaining *wild* ramified places, while Theorem 7.6 handles the tame ones, the resulting map Ψ_D from irreducible genuine representations of the cover to $\mathrm{Irr}(D_v^\times)$ can be composed, subject to central-character and normalization checks, with

$$\mathrm{Irr}(D_v^\times) \xrightarrow{\mathrm{JL}} \mathrm{Irr}_{\mathrm{ds}}(\mathrm{GL}_r(F)) \xleftarrow{\mathrm{MC}} \mathrm{Irr}_{\mathrm{gen,ds}}(\widetilde{\mathrm{GL}}_r(F)),$$

where MC is Zou's corrected form of the Flicker–Kazhdan correspondence, with the required restriction that the linear central character be trivial on μ_n. At tame places this composition gives the proved conclusion of Corollary 7.8 in the stated range of the external Flicker–Kazhdan theorem; outside that range it remains conditional on an applicable extension of the external input. This would establish a local metaplectic Jacquet–Langlands correspondence. The fact that local supercuspidals are easier to isolate spectrally does not eliminate the wild-place integrality obstruction for all representations.

Similarly, removing the non-cancellation hypothesis (H'_D) of Part II, Definition 6.4, would require showing that a prescribed packet has only finitely many S-types, or some other condition contradicting the cancellation criterion from Part II. The original planning memo's argument through strong multiplicity one is a plausible *conditional* route, but its spectral separation and finiteness arguments have not yet been established.

Removal of the extra cuspidal place w_0, or of the converse hypothesis (H') (Part II, Corollary 6.8), remains a different problem. None of these goals is asserted as a theorem in the present draft.

13 Selected terminology and notation

These brief definitions collect the notions used in the character and Kottwitz arguments; standard background may be found in [4, 5, 1, 2].

Adams operation ψ^n
: The λ-ring operation on the representation ring satisfying $\chi_{\psi^n V}(g) = \chi_V(g^n)$; it can produce a virtual character rather than an actual representation.

Equivariant K-theory
: At a point, $K_Q^0(\mathrm{pt}) = R(Q)$, the representation ring of a finite group Q. Exterior powers give its λ-operations.

Virtual character
: An integral linear combination of irreducible characters.

Hall π-subgroup
: A subgroup whose order involves only primes in π and whose index is prime to every prime in π; in the tame proof it separates the central n-part of a finite quotient.

Good element
: A regular lift commuting with the inverse image of its ordinary centralizer. Genuine invariant functions vanish on nongood lifts.

Normalized orbital integral
: Here $J(t, f) = \Delta(t)O(t, f)$, where Δ is the square root of the Weyl discriminant and quotient Haar measures are specified.

Transforms $L_D, \mathcal{O}_D, \mathcal{B}$
Respectively the unitary invariant class-function transform, the forward test-function transfer, and its inverse modulo orbital-integral kernels; see Definitions 4.1 and Theorem 9.1.

Stable/rational conjugacy
Conjugacy over an algebraic closure (with descent) versus conjugacy by the rational points $G(K)$; one stable class may contain several rational classes for a general reductive group.

Inner form A twist of a reductive group by inner automorphisms; $\mathrm{GL}_1(D)$ is an inner form of GL_r.

Induced torus
In this paper $T = \mathrm{Res}_{E/K}\,\mathbb{G}_m$, with $T(K) = E^\times$ and $H^1(K, T) = 1$ by Hilbert 90.

Kottwitz torus obstruction
A global cohomological obstruction to transferring an admissible torus to an inner form despite local embeddings, discussed in [1, §9.5].

Relative Tamagawa number
The ratio $\tau_1(G) = \tau(G)/\tau(G_{\mathrm{sc}})$; it is not a local probability Haar measure.

Brauer class The class $[D] \in \mathrm{Br}(K)$ of a central simple algebra; E embeds maximally in D exactly when $D \otimes_K E$ splits.

Scalar power quotient
$\mathcal{Q}(E/K, n) = E^\times/(K^\times E^{\times n})$; it measures global scalar-modulo-nth-power classes.

Kummer and Sha^1
The exact sequence $1 \to T[n] \to T \xrightarrow{n} T \to 1$ relates powers to Galois cohomology. $\mathrm{Sha}^1(K, T[n])$ is the kernel of localization of $H^1(K, T[n])$ at every place.

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